YOUSSEF HESHAM NASR

R&D Mechanical Design Engineer | Gym & Fitness Equipment | Certified SOLIDWORKS Professional (CSWP)

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PROFESSIONAL SUMMARY

R&D Mechanical Design Engineer specializing in biomechanics-driven design of commercial and home fitness equipment. Takes machines from concept through 3D design, FEA validation and DFM to production-ready manufacturing files (DXF, IGES, fabrication drawings, BOMs) and patent drawings. Certified SOLIDWORKS Professional (CSWP + Sheet Metal) with 25+ delivered projects for international clients, hands-on CNC laser/plasma nesting experience, and machines manufactured directly from own production files.

CORE SKILLS

- **CAD & Tools:** SOLIDWORKS (Part, Assembly, Weldments, Sheet Metal, Simulation), AutoCAD, KeyShot
- **Design:** Mechanical Design, Mechanism Design, Biomechanics, Product Development, Prototyping, Reverse Engineering
- **Analysis:** FEA / Stress Analysis, Equivalent Strain Studies, Structural Validation, Stability & Tip-Over Analysis
- **Manufacturing:** Design for Manufacturing (DFM), Design for Assembly (DFA), Sheet Metal, Weldments, Tube/Pipe Frames, Flat Patterns, CNC Laser & Plasma Cutting, Nesting, G-code, Powder Coating
- **Documentation:** 2D Fabrication Drawings, Bills of Materials (BOM), DXF, IGES, STEP, PDF Drawing Packages, Patent Drawings
- **Industries:** Commercial Strength Equipment, Home Gym Equipment, Rehabilitation & Medical Devices, Baby Mobility & Strollers

PROFESSIONAL EXPERIENCE

Mechanical Design Engineer, Gym Equipment Division **Aug 2025 - Present**

ISM LTD

- Lead R&D on gym equipment design and biomechanics, integrating complex mechanisms into unconventional machine concepts.
- Prepare complete manufacturing files and fabrication drawings for production.
- Draft patent documentation and drawings supporting product approval for Saudi Arabia, ISO and the wider Middle East.

Contracted R&D Mechanical Design Engineer **Jun 2025 - Present**

Rebuilt Company

- Lead engineering and DFM of commercial fitness equipment, including heavy-duty power racks and plate-loaded hip thrust machines.
- Designed a power trainer rack with an integrated counterweight-assist system and a hip thrust machine built for commercial duty cycles.
- Validate structural integrity of welded frames through FEA equivalent strain studies.
- Optimize sheet-metal nesting for laser-cut manufacturing and create production-ready sheet-metal brackets with full flat patterns.

Freelance Mechanical Design Engineer **Aug 2024 - Present**

Fiverr (Global)

- Delivered 25+ engineering projects for international clients: 3D models, FEA simulations and DFM optimization across gym, rehabilitation, home fitness, medical and consumer products.
- Since Oct 2026, also manufacture client-designed gym machines (laser cutting, welding, powder coating, assembly) and ship them worldwide as a separate service.
- Produce photorealistic KeyShot renders and animations for marketing and design sign-off.

CNC Nesting & Reverse Engineering **Apr 2025 - Aug 2025**

El-Nesr For Sheet Metal Cutting

- Reverse engineered real-world sheet-metal parts by measuring, drafting clean CAD geometry and exporting production-ready DXF files.
- Owned the nesting pipeline on CNC plasma and laser machines: layout optimization, G-code adjustment and parameter tuning (cutting speed, kerf, pierce).
- Reduced material cost through aggressive nesting strategies that minimized scrap and maximized sheet yield.

SELECTED PROJECTS

- **Pediatric Rehabilitation Leg Press (designed and built):** Designed around one child's body measurements (3 kg working load, +4 kg optional); produced DXF and IGES files, then manufactured it: laser-cut, welded, powder-coated and assembled.
- **Strength Equipment Portfolio:** Ultimate Leg Machine, Linear and Iso-Lateral Leg Press, Preacher Curl Machine, Functional Trainer / Cable Tower, Multifunctional Trainer, Home Cable Machine, plus dumbbell and weight-plate product models.
- **Baby Mobility & Consumer Products:** 3-in-1 convertible platform combining stroller, tricycle and car seat; collapsible silicone bathtub with living-hinge geometry.
- **Medical Devices:** OMNISTEP-B1 hospital bed with an integrated lower-limb physiotherapy carriage.
- **Reverse Engineering:** Rebuilt a production-ready CAD model of a vanity mirror stand from physical reference geometry.

CERTIFICATIONS

- Certified SOLIDWORKS Professional (CSWP), Dassault Systemes
- CSWP - Sheet Metal, Dassault Systemes
- Certified SOLIDWORKS Associate (CSWA), Dassault Systemes
- ITIDA Gigs Freelance Program, ITIDA & EYouth, 2024
- Career Directions Program, ITIDA, TIEC & Creativa Tanta Innovation Hub, 2024

EDUCATION

B.Sc. Mechatronics Engineering **Expected 2027**

Tanta University, *Egypt*

- Relevant coursework: mechanical design, kinematics, material science, structural analysis, electronics and control systems.

LANGUAGES

Arabic (Native) | English (Professional Working Proficiency)